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Effects of predation and nutrient enrichment on the success and microbiome of a foundational coral

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Abstract. By inflicting damage to prey tissues, consumer species may increase stress in prey hosts and reduce overall fitness (i.e., primary effects, such as growth or reproduction) or cause secondary effects by affecting prey interactions with other species such as microbes. However, little is known about how abiotic conditions affect the outcomes of these biotic interactions. In coral reef communities, both nutrient enrichment and predation have been linked to reduced fitness and disease facilitation in corals, yet no study to date has tested their combined effects on corals or their associated microbial communities (i.e., microbiomes). Here, we assess the effects of grazing by a prevalent coral predator (the short coral snail, *Coralliophila abbreviata*) and nutrient enrichment on staghorn coral, *Acropora cervicornis*, and its microbiomes using a factorial experiment and high-throughput DNA sequencing. We found that predation, but not nutrients, significantly reduced coral growth and increased mortality, tissue loss, and turf algae colonization. Partial predation and nutrient enrichment both independently altered coral microbiomes such that one bacterial genus came to dominate the microbial community. Nutrient-enriched corals were associated with significant increases in *Rickettsia*-like organisms, which are currently one of several microbial groups being investigated as a disease agent in this coral species. However, we found no effects of nutrient enrichment on coral health, disease, or their predators. This research suggests that in the several months following coral transplantation (i.e., restoration) or disturbance (i.e., recovery), Caribbean acroporid corals appear to be highly susceptible to negative effects caused by predators, but not or not yet susceptible to nutrient enrichment despite changes to their microbial communities.

Key words: community ecology; coral reef; corallivore; gastropod; microbial interactions.

Introduction

Across diverse ecosystems, changing environmental conditions are affecting ecological interactions, such as predator-prey interactions, and their outcomes (Githeko et al. 2000, Silliman et al. 2013). In addition to the primary effects of complete predation (e.g., mortality) or partial predation on prey (e.g., reduced fitness), consumers may also inflict secondary effects by altering associations with microbial communities. When these changes favor or facilitate pathogenic microbes, whole ecological communities can be affected if disease hosts are also foundation species. For example, pine trees in the southwest United States are commonly grazed on by scolytid pine bark beetles that facilitate fungus in their grazing wounds (Paine et al. 1997). During intense drought, major die-off events of pine trees have occurred as drought-stressed trees succumbed to fungal infections and beetle grazing (Raffa et al. 2008). Similarly, thousands of hectares of eastern US salt marshes have been lost due to a combination of drought and fungal infection facilitated by grazing periwinkle snails (Silliman et al. 2005). These interactions between abiotic conditions, grazers, and opportunistic microbial infections not only lead to declines in host foundation species but may present challenges to the successful restoration of critical habitats.

In coral reef ecosystems, nutrient enrichment is an environmental factor often studied in connection to coral disease. For instance, nutrient enrichment has been shown to increase the progression of yellow band disease (Bruno et al. 2003) and black band disease (Voss and Richardson 2006), the severity of aspergillosis in sea fans (Bruno et al. 2003), and the prevalence of dark spots disease (Vega Thurber et al. 2014). Excess nutrients can also increase virulence genes expressed by microbes living on the coral surface mucosal layer (Vega Thurber et al. 2009), ultimately affecting the microbial communities believed to help maintain coral fitness (Rosenberg et al. 2007) and protect corals against invading pathogens (Ritchie 2006). Most recently, nutrients have been found to interact with coral predation, leading to increased mortality after

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corals received grazing wounds by parrotfish (Zaneveld et al. 2016). These results suggest that changing abiotic conditions may interact with coral predators to affect microbes and disease.

Like other grazers, coral predators may be able to alter coral-associated microbial communities by creating open feeding wounds. Several studies have found corals developed disease signs when in the presence of diseased corals and when open wounds were created by either coral predators or simulated damage (Aeby and Santavy 2006, Page and Willis 2008). In the Caribbean, predation by the corallivorous gastropod *Coralliophila abbreviata* (the short coral snail) can increase transmission of disease symptoms similar to white band disease (Williams and Miller 2005, Gignoux-Wolfsohn et al. 2012) between healthy and diseased colonies of corals of the genus *Acropora*. In addition, grazers and nutrients may interact to facilitate disease through increased rates or intensity of grazing, for example by increasing grazing rates on nutrient-enriched tissues (marsh grasses: Silliman and Zieman 2001, terrestrial plants: Throop and Lerdau 2004), or increasing grazer population density (e.g., crown-of-thorns starfish: Brodie et al. 2005). However, little is known about the primary effects of these factors on corals in combination or their secondary effects on altering specific microbial taxa or overall microbial communities.

Here, we explore these interactions in coral reef ecosystems using a fully factorial field experiment on the Caribbean staghorn coral, *Acropora cervicornis*, assessing primary effects on corals (e.g., survivorship, growth) and secondary effects for coral microbiomes. Historically one of the most abundant reef-building corals in the Caribbean, this species is currently listed as critically endangered under the IUCN Red List (Aronson et al. 2008) and threatened under the US Endangered Species Act (Hogarth 2006) after disease caused extensive population mortality in the 1980s (Aronson and Precht 2001). Although efforts to restore *A. cervicornis* populations are currently underway across the Caribbean, acroporid corals continue to succumb to predation, disease, and harmful abiotic conditions (Williams and Miller 2012). Both predation by *C. abbreviata* and nutrient enrichment have been cited as major concerns for practitioners engaged in coral reef restoration (Miller 2001, Johnson et al. 2011, Young et al. 2012, National Marine Fisheries Service 2014). Given these restoration priorities, determining how these top-down and bottom-up factors affect the success of restoration efforts is both timely and important as coral decline continues.

Methods

Survey of corallivory on staghorn colonies

To determine the prevalence of predation on *Acropora cervicornis* by *Coralliophila abbreviata* in natural settings, we conducted surveys of both natural and restored colonies across four coral reefs in the Florida Keys National Marine Sanctuary (Molasses Reef: N 25 0.579, W 80 22.471; Pickles Reef: N 24 59.170, W 80 24.940; Conch Reef: N 24 57.110, W 80 27.564; American Shoals: N 24 31.568, W 81 31.383). At each site, we surveyed all *A. cervicornis* colonies found in 15–20, 2 × 20 m belt transects and recorded the density of *C. abbreviata* per colony, as well as colony height, length, and width to the nearest cm.

Predation × nutrient experiment

To assess how predation and nutrients affect corals and their microbiomes, we exposed colonies of *A. cervicornis* to a fully factorial field experiment from June to August 2013. Due to the threatened status of this species, this experiment could not be conducted in natural reef settings or with already-restored colonies. Thus, we conducted this experiment in the Coral Restoration Foundation's coral nursery in Key Largo, FL, USA in an open sand flat (30-ft depth) using corals grown through established propagation techniques (Johnson et al. 2011). We collected coral fragments using wire cutters from visually healthy (e.g., non-diseased) colonies of two genotypes (Upper Keys 2, "U-2"; Upper Keys 8, "U-8") due to possible genotypic variation in disease susceptibility. All coral fragments were ~10 cm long to approximate coral transplant sizes used in restoration activities and due to limits in using nursery corals. Using All Fix Marine Epoxy, we secured each coral fragment to an individual structure consisting of a concrete slab with attached vertical PVC pipes, with corals attached to the top of one PVC pipe. Structures were placed 3 m apart in all directions and were cleaned of all fouling organisms prior to experimentation.

After a 1-week acclimation period, we assigned corals to one of four treatments: (1) snail + nutrient; (2) snail-only; (3) nutrient-only; and (4) no snails or nutrients (control). Each treatment consisted of 10 replicates per genotype for a total of 80 colonies. To ensure variation in environmental conditions did not affect treatment results, we evenly distributed genotypes and treatments throughout the experimental site. For nutrient addition treatments, diffusion tubes were secured to PVC tubes ~5 cm below coral colonies and replaced every 2–3 weeks. Diffusion tubes consisted of a 6-inch long × 1-inch wide PVC pipe with 16, 3/8-inch holes drilled into the side and covered in mesh screen. Tubes were filled with 70 g Osmocote 19-6-12 Smart-Release plant fertilizer pellets following the methods of Worm et al. (2000). To ensure nutrient diffusion tubes were effective in releasing nutrients, we analyzed inorganic nitrogen and phosphorus spectrophotometrically from water samples taken within ~3 cm of 5 control corals and ~1 cm from 5 nutrient diffusion tubes a few days after diffusion tubes were established (UV-2450, Shimadzu; Southeast Environmental Research Center, Miami, Florida, USA). Inorganic nutrients were significantly higher in water

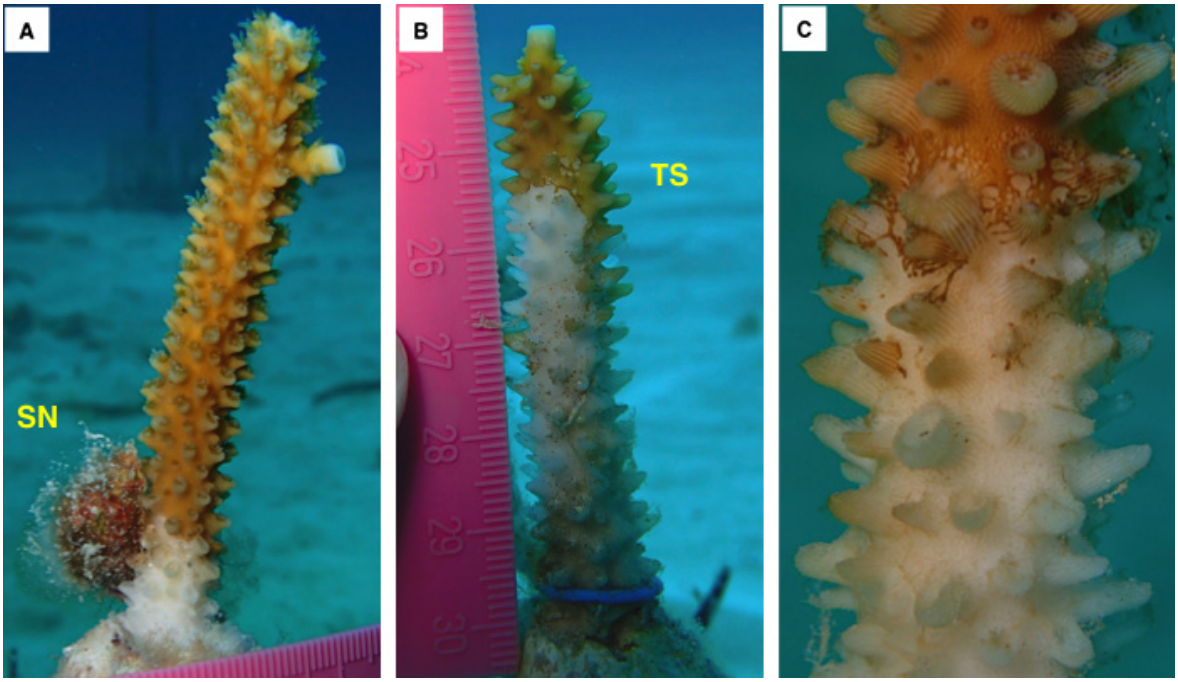

FIG. 1. (A) *Coralliophila abbreviata* on *Acropora cervicornis* ("SN") creating feeding scars identified as white skeleton at base of coral, (B) tissue sloughing ("TS") off *Acropora cervicornis* after snail removal, and (C) tissue sloughing leading to rapid coral mortality.

immediately adjacent to the nutrient tubes (N = 1.474 ppm; P = 0.4211 ppm) relative to controls (N = 0.007 ppm, P_N = 0.012; P = 0.007 ppm, P_P = 0.006). Though samples were not directly next to corals, we believe this suggests corals received heightened levels of nutrients throughout the experiment.

For predator addition treatments, we collected *C. abbreviata* snails from non-diseased corals on nearby corals reefs and starved snails in an aquarium for one week to ensure that any potential pathogens were not transmitted through fecal matter. We placed two snails at the base of each snail-only and snail + nutrient coral, which approximated naturally high snail densities on surveyed *A. cervicornis* colonies harboring *C. abbreviata* when corrected for colony height (see *Results*). Every 2–3 days, we examined corals for predation scars indicated by white coral skeleton (Fig. 1A) and removed snails from colonies when a large portion of tissue (~30% of colony surface area) was consumed. We recorded colony mortality, tissue loss, and disease signs (e.g., tissue sloughing; Fig. 1B, C) for all treatments every 2–3 days during the study. To assess coral growth, we measured the total linear extension (e.g., additive length of all branches, TLE) of corals at the beginning and end of the experiment. After observing that turf algae began colonizing the bare coral skeleton left by snail feeding scars, we additionally measured algal colonization on coral colonies at the end of the experiment. This was done by visually estimating the percent cover of algae within snail feeding scars and on live coral tissue to the nearest 5%.

Coral microbiome collections and processing

To determine the effects of nutrients and predators on coral microbiomes, we randomly selected five colonies from each treatment (including both genotypes) for sampling at the end of the experiment. Part of the main branches of corals were clipped and placed in 50 mL centrifuge tubes and frozen until analysis. Feeding scars were included in samples for all snail addition treatments. We did not sample corals with disease signs as rapid mortality occurred before sampling could be conducted. We used high-throughput 16S amplicon sequencing to assess the bacterial and archaeal communities associated with each coral. To do this, we isolated and homogenized coral tissue from both the tip and the base of each coral fragment with a WaterPik and sterile Milli-Q water. Total DNA was extracted from 200 μL of the resulting tissue slurry with a MoBio PowerSoil kit according to manufacturer's protocol. Barcoded primers 515F and 806R and ToughMix PCR (Quanta BioSciences, Gaithersburg, Maryland, USA) were used to amplify and tag the V4 region of the bacterial and archaeal 16S rRNA gene. Samples were amplified in triplicate, with each technical replicate given a different combination of barcodes. Triplicates were pooled, cleaned, and quantified before all samples were combined in equimolar ratios. Multiplexed libraries were submitted to Oregon State University's Center for Genome Research and Biocomputing (CGRB) for sequencing on the Illumina MiSeq platform.

MiSeq data was processed in the QIIME pipeline using MacQIIME v1.9.1 Reads were first separated by barcode tags and processed for quality by trimming low-quality bases, discarding short reads, and filtering chimeras with USEARCH 6.1544 default parameters (Caporaso et. al 2010). Reads were clustered into operational taxonomic units (OTUs) using pick_open_reference_otus.py a 97% similarity threshold, and USEARCH v6.1544. OTUs were assigned taxonomy using UCLUST and the Greengenes 13_10 97% reference database. Mitochondrial and chloroplast OTUs were filtered from the data based on taxonomy. Data from the tip and base of each individual coral were then pooled informatically to create the "raw" OTU table.

Statistical analysis

Coral fragments that went missing due to roaming turtles and storms were removed from statistical analyses (snail + nutrient: $n = 3$; snail-only: $n = 2$; nutrient-only: $n = 0$; control: $n = 3$). Colony mortality was defined as a total loss of living tissue from the coral fragment and was analyzed using a generalized linear model (GLM) with a binomial distribution due to uneven sample sizes between treatments. Tissue loss was only observed on snail addition colonies during our experiment; therefore, we examined the effects of nutrients on tissue predation. To do this, we estimated the area of tissue predation using the equation for the area of an ellipse ($A = \pi ab$) to approximate the natural shape of *C. abbreviata* feeding scars. We used measurements of the maximum height and width of scars taken at the end of the experiment or before tissue sloughing was observed on corals. When predation scars extended across the circumference of the coral, we averaged the maximum and minimum scar height and multiplied it by the circumference measured at the base of each individual coral. Tissue loss data were square root transformed to meet the assumptions of parametric statistics and Analysis of Variance (ANOVA) was used to analyze the effect of nutrient enrichment and coral genotype on tissue predation. To examine treatment effects on coral growth, corals that suffered full colony mortality or were broken and reattached in the middle of the experiment were excluded from this analysis. Growth data were analyzed using a GLM (nutrients × predation + genotype). The percent cover of filamentous turf algae on coral colonies was analyzed using beta regression with the *betareg* package in R (Cribari-Neto and Zeileis 2010), as beta regression is used to model continuous sample proportions. All analyses were conducted in R version 3.2.1 (R Core Team 2015).

To assess treatment effects on coral microbiomes, we performed Principal Coordinates Analysis (PCoA) using QIIME's beta_diversity_through_plots.py. We calculated weighted UniFrac distances and principal coordinates after rarefying the raw OTU table to a uniform depth of 43,000 reads per sample. Data were loaded into R using the *Phyloseq* package v1.12.2 (McMurdie and Holmes 2014, R Core Team 2015). We used Levene's tests and ANOVAs to assess treatment affects for PCoA axes using the *car* package v2.0.25 in R. Correlations between OTUs and PCoA axes were determined with QIIME using the rarefied table and observation_metadata_correlation.py. Correlations with family-level taxa were assessed by pooling these OTUs with summarize_taxa.py.

Prior to differential abundance analysis, OTUs present in fewer than four samples (>0.5n) were removed from the pre-rarefaction OTU table. Additionally, we used only OTUs that were found in at least 25% of samples and with counts of at least 10 (out of 30,000 reads). R packages *Phyloseq* and *DESeq2* v1.8.2 (Love et al. 2014) were used to perform Wald tests for differential abundance of individual OTUs using the model formula ~nutrients + snails + nutrients:snails. The effect of genotype on microbial communities was not explored due to limited sample sizes. Correlation of OTUs with coral growth rate, tissue loss, and algal colonization was assessed in QIIME by rarefying the filtered OTU table to 30,000 reads per sample and using observation_metadata_correlation.py. All reported *P*-values are FDR-corrected.

Results

Surveys of predation on staghorn corals

A total of 198 *Acropora cervicornis* colonies were recorded across the four reefs surveyed. Of these, 15.2% ($n = 30$) were found harboring *Coralliophila abbreviata*. Snail density ranged from 0 to 7 snails per colony and averaged 0.35 ± 1.01 snails per colony (mean ± SD) across all colonies surveyed, but increased to 2.30 ± 1.51 snails on colonies harboring snails (range: 1–7 snails per colony). When standardizing *C. abbreviata* density by colony branch length, we found snail density averaged 0.01 ± 0.01 snails per cm. Because *C. abbreviata* do not often climb onto secondary branches of *A. cervicornis* (E. C. Shaver, *personal observation*), we also calculated snail density per colony height (e.g., of the primary branch) and found *C. abbreviata* averaged 0.14 ± 0.08 individuals per cm.

Effects of nutrients and predation on growth, tissue loss, and mortality

Predation by *Coralliophila abbreviata* lead to the death of 33% of snail addition colonies by the end of the experiment ($n = 11$; Fig. 2A); in contrast, no nutrient-only or control colonies suffered full mortality (GLM, $P_{snail} < 0.0001$, $\chi^2 = 16.9$). Snails were removed from 10 corals in snail-only treatments and five corals in snail + nutrient treatments after large predation scars were made and 60% of these corals suffered full mortality after snails were removed. These corals included 29.4 ± 11.4% of snail + nutrient ($n = 5$) and 33.3 ± 12.1% ($n = 6$) of snail-only colonies. When comparing only snail-addition corals, we found no effect of nutrient enrichment

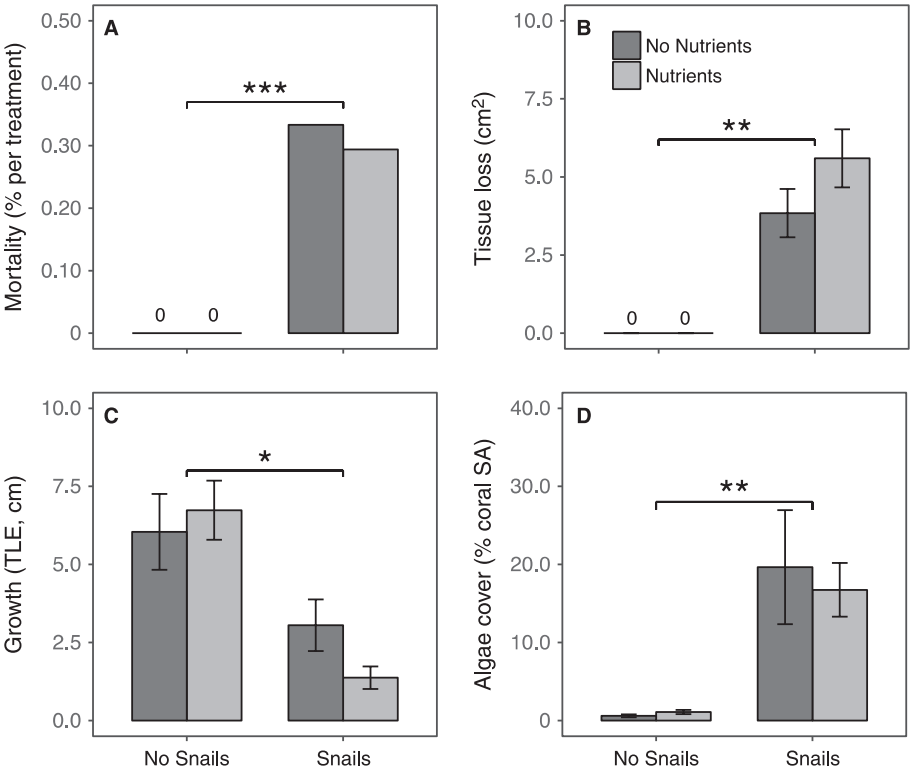

FIG. 2. Effects of predation by *Coralliophila abbreviata* and nutrient enrichment on (A) full colony mortality (as the proportion of colonies per treatment), (B) tissue loss per coral colony, (C) growth as a measure total linear extension of all branches, and (D) turf algae colonization on colonies (as the percent of colony surface area). Error bars represent standard error of the mean. Significance values are denoted as such: "***" for $P < 0.0001$, "**" for $P = 0.0001$–0.001, and "*" for $P = 0.001$–0.05.

($P = 0.797$, $t = -0.257$). We additionally found no effect of genotype ($P = 0.6314$, $z = 0.48$). We observed disease symptoms similar to those reported for rapid tissue loss syndrome (e.g., rapid death, tissue sloughing) on all corals that suffered full mortality (Williams and Miller 2005, Miller et al. 2014*b*).

Coral tissue loss increased from 0% without snails to approximately ~30% of live tissue surface area with snail predation (GLM, $P_{snail} = 0.0005$, $\chi^2 = 12.4$). We found no effects of nutrient enrichment (with nutrients: 5.60 ± 3.94 cm^2, without nutrients: 3.84 ± 3.37 cm^2; $P = 0.192$, $F_{1,34} = 1.773$) or genotype ($P = 0.621$, $F_{1,34} = 0.248$) on tissue predation (Fig. 2B). The growth of corals with snail predation was reduced by 64% relative to corals without predation ($P = 0.0107$, $t = -2.687$) (Fig. 2C). However, growth was most significantly affected by genotype, where U-8 corals grew 71% more than U-2 corals (U-8: 5.75 ± 0.74 cm, U-2: 3.35 ± 0.50 cm; $P = 0.0020$, $t = 3.328$). We found no effect of nutrients ($P = 0.5281$, $t = 0.637$) or the interaction between predation and nutrients ($P = 0.1881$, $t = -1.341$). While nutrient-only and control corals grew the most during this experiment (N: 6.74 ± 0.95 cm; C: 6.04 ± 1.21 cm), growth of snail-only and snail + nutrient corals were reduced to 3.06 ± 0.83 cm and 1.38 ± 1.02 cm, respectively.

The colonization of turf algae on coral colonies was also significantly affected by predation (Fig. 2D; $P = 0.0026$, $z = 3.009$), but not by nutrients ($P = 0.7604$, $z = 0.0305$), genotype ($P = 0.9793$, $z = -0.026$), or the interaction between predation and nutrients ($P = 0.8845$, $z = 0.145$). Algae covered 18.1 ± 3.4% of total colony surface area in corals subjected to predation, while algal colonization on nutrient-only and control colonies was rare (~0.01% algal cover). The majority of algal colonization occurred within snail predation scars (92.0 ± 3.3%), covering on average 57.4 ± 6.7% of snail scars, while little colonization occurred on live coral tissue (1.1 ± 0.22% algal cover).

Effects of nutrients and predators on A. cervicornis *microbiomes*

Overall, 64.84% of the variation observed in coral microbiomes was explained by the first two PCoA axes (Appendix S1: Fig. S1). Variation in PC1 was significantly greater in nutrient-enriched corals (Levene test, $P = 0.017$; Fig. 3A), and was correlated with the abundance of OTU 150441 (Rickettsiales; $R^2 = 0.992$, $P < 0.0001$; Fig. 3B). The relative abundance of this Rickettsiales OTU increased from at most 11.4% of the bacterial community without nutrients to at most 87.9%

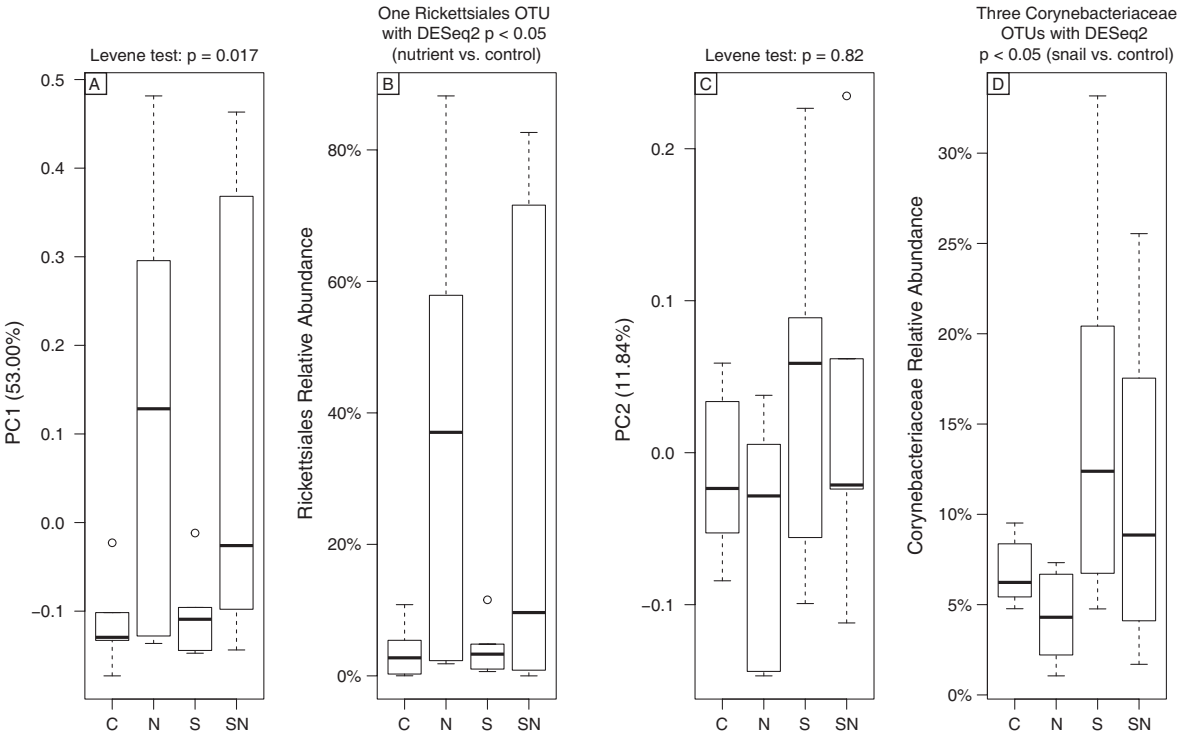

FIG. 3. Results of Principle Coordinates Analysis (PCoA) for each of the four treatments examined in the field experiment (C = control/no snails or nutrients; N = nutrient-only; S = snail-only; SN = snail + nutrient). The variance in PC1 was significantly greater in nutrient-enriched samples (A), which was strongly correlated with the abundance of a single OTU in the order Rickettsiales (B). The variance in PC2 was not significantly affected by either treatment (C), however predation by *Coralliophila abbreviata* was strongly correlated with the abundance of five OTUs in the family Corynebacteriaceae (D).

with nutrient enrichment (Wald test, $P = 0.030$; Fig. 3B). One other OTU (#12574, *Actinomyces*) increased due to nutrients, where it was at most 0.8% in control corals but increased to at most 1.1% of the community in enriched corals (Wald test, $P = 0.04$) (Table 1).

The variance of PC2 was not significantly different among treatments (Levene test, $P = 0.82$). However, PC2 was correlated with the abundance of the genus Corynebacteriaceae ($R^2 = 0.708$, $P < 0.0001$; Fig. 3C), which had three OTUs significantly increased in snail treatments (Wald test, all $P < 0.03$). The most abundant of these, OTU 1015518, was at most 2.2% of the community in control and nutrient-only corals, but increased to at most 10.0% in snail-only and snail + nutrient corals. All Corynebacteriaceae together comprised at most 9.5% without snails and at most 33.2% with snails (Fig. 3D). Two *Actinomyces* OTUs were also more abundant in snail treatments (Wald test, both $P < 0.03$). No OTUs had significant interactions between the treatments (Table 1).

Although no bacteria were correlated with coral growth across treatments, we found a strong negative correlation between coral growth and OTU 150441 (Rickettsiales) in the nutrient-only treatment ($R^2 = 0.999$; $P < 0.001$; Appendix S1: Fig. S2). In contrast, we found no

TABLE 1. Effects of predation by the short coral snail, *Coralliophila abbreviata*, and nutrient enrichment on the differential abundance of microbial operational taxonomic units (OTUs) using DESeq2.

Sig. factor	OTU #	Control %	Treatment %	*P*-value	Taxonomy
Nutrients	150441	4.5	35.3	0.03	*Rickettsiales*
Nutrients	12574	0	0.19	0.04	*Actinomyces*
Snails	377613	0.04	1.1	0.01	*Corynebacterium*
Snails	1015518	0.09	3.0	0.02	*Corynebacterium*
Snails	410908	0.007	0.17	0.03	*Corynebacterium*
Snails	4334479	0	0.09	0.01	*Actinomyces*
Snails	12574	0	0.15	0.03	*Actinomyces*

TABLE 2. Significant correlations among microbial OTUs in coral microbiomes and metrics of coral health. *P*-values are false discovery rate (FDR) corrected which depicts the number of type 1 errors detected by this analysis.

Sig. factor	Within treatment	OTU	R^2	*P*-value	Taxonomy
Coral growth	Nutrients-only	150441	−0.9986	0.0030	Rickettsiales
Algal colonization	Snail treatments	4448456	0.8295	0.0310	*Streptococcus*
Coral tissue loss	Snail + Nutrient	136025	0.9610	0.0196	*Ralstonia*
Coral tissue loss	Snail + Nutrient	254888	0.9551	0.0196	Comamonadaceae
Coral tissue loss	Snail + Nutrient	66026	0.9532	0.0196	*Vibrio*

relationship between growth and OTU 150441 in the control, snail-only, or snail + nutrient treatments (all $P > 0.47$) (Table 2; Appendix S1: Fig. S2). Across both snail treatments, OTU 448456 (*Streptococcus* sp.) was positively correlated with algal colonization ($R^2 = 0.830$, $P = 0.03$). In the snail + nutrient treatment, three OTUs belonging to the taxa *Ralstonia*, Commamonadaceae, and *Vibrio* were positively correlated with tissue loss (all $R^2 > 0.953$, $P < 0.02$). No other OTUs were correlated with coral growth, tissue loss, or algal colonization (Table 2). Overall microbial diversity did not vary across treatments, likely due to low sample sizes (Appendix S1: Fig. S3).

DISCUSSION

In this study, coral predators had significant primary effects on the success of *Acropora cervicornis* colonies, including survivorship and growth, while both predation and nutrient enrichment independently altered coral-associated microbial communities. Each factor altered coral microbiomes such that particular bacteria genera came to dominate microbial communities, and several microbial OTUs strongly correlated with coral growth, algal colonization, and tissue loss. While nutrient enrichment had the strongest effect on coral microbiomes, we found no effects of nutrients on coral health or disease in our experiment. These results suggest that small acroporid corals may be highly susceptible to negative effects caused by predators in the months following restoration or disturbance, but are not or not yet susceptible to nutrient enrichment despite its effects on coral microbial communities. Further research is required to determine how responses of coral microbiomes to external forces ultimately affect coral health.

We found that partial, sub-lethal predation on healthy corals could lead to rapid colony mortality when grazing wounds became large (here, ~30% of coral surface area). In our experiment, corals with partial tissue predation by *Coralliophila abbreviata* were nearly five times more likely to suffer complete mortality than corals without predators. While the densities of corallivorous snails used in our experimental predator additions were on the high end of naturally observed densities found in our survey (experiment: 0.2 snails/cm of coral; survey: 0.14 ± 0.08 snails/cm of coral), our results are similar to other studies showing particularly high feeding rates on acroporid corals by *C. abbreviata* (Brawley and Adey 1982, Knowlton et al. 1990, Baums et al. 2003*b*). Other studies have also found continued loss of *A. cervicornis* tissue post predation by the bearded fireworm (*Hermodice carunculata*, Miller et al. 2014*a*); however, we observed complete coral colony mortality after predatory snails were removed, with tissue sloughing similar to disease symptoms observed in the field (Williams and Miller 2005). In our study, we allowed tissue sloughing to continue in order to observe the frequency with which partial predation lead to full colony mortality and were therefore unable to obtain microbial samples from corals with apparent disease signs.

Predation also reduced the growth of experimental corals by nearly 65% relative to control colonies, a result similar to other studies suggesting that predation increases energetic demands and may reduce overall coral fitness (Oren et al. 1998, Rotjan and Lewis 2008). For instance, predation by butterflyfish reduced the growth of the coral *Montipora verrucosa* by 55% relative to caged colonies protected from predators (Cox 1986). Results of our survey and others suggest the impacts of predation may be limited to a small portion of Caribbean acroporid populations (here, 15.2% of *A. cervicornis*; Schopmeyer and Lirman 2015: 7.9, % of *A. cervicornis*; Miller et al. 2008: ~5% of *A. palmata*). However, top-down control of corals by predators may increase as prey (coral) decline and predators aggregate on remaining colonies, a process that has been documented in several other ecosystems including coral reefs (Silliman et al. 2013). Finally, corals subjected to predation were six times more likely to become colonized by turf algae due to predation scars. This is likely due to the feeding characteristics of this and other corallivorous snails, which feed continuously on coral tissue and create large feeding scars, inhibiting tissue repair and allowing for the colonization of turf algae (Van Woesik 1998). Although we were unable to quantify the effects of predator-facilitated algae on coral fitness or microbiomes, recent research shows that contact with algae can alter microbial communities leading to coral tissue loss, disease, and mortality (Zaneveld et al. 2016). We did, however, find a positive correlation between algal colonization and an OTU in the genus *Streptococcus*.

These results suggest that coral predators may have strong effects on small *A. cervicornis* colonies being restored, potentially limiting the recovery of this species. While the manual removal of *C. abbreviata* from restoration sites is cited as a priority for improving acroporid restoration success (National Marine Fisheries Service

2014), increased understanding of natural mechanisms controlling their populations are also needed. For instance, overfishing of reef predators has been suspected of causing corallivore outbreaks in the Indo-Pacific (Sweatman 2008), Hawaii (Gochfeld and Aeby 1997), and Kenya (McClanahan 1997), and manipulative removals of pufferfish in Florida led to a 19-fold increase in corallivorous gastropods that prey on soft corals (Burkepile and Hay 2007). Although the extent of top-down control of *C. abbreviata* populations remain unknown, a trophic cascade may contribute to their regulation and their consumer pressure on habitat-building corals as these snails may be consumed by high-value commercial species such as the Caribbean spiny lobster (*Panulirus argus*), hogfish (family: Labridae), or triggerfish (family: Balistidae) (as proposed in Baums et al. 2003*a*).

In contrast, we did not find any effects of nutrient enrichment on coral tissue loss, mortality, growth, or algal colonization. The lack of nutrient effects may have been due to the short duration of our experiment, particularly as turf algae can overgrow dead surfaces of coral colonies under nutrient enrichment conditions (McClanahan et al. 2003, Vermeij et al. 2010). The relationship between nutrients and coral growth can also vary across nutrient concentrations (Gil 2013, Shantz and Burkepile 2014), and therefore results of our presence/absence nutrient treatments also limit our interpretation. Additionally, the levels of nutrients exposed to corals also likely affected our results; however, we did not attempt to determine these levels through seawater sampling as nutrient concentrations often vary with changes in temperature, flow, and other environment variables. Nutrient exposure in our methods closely followed other studies (here: ~5.0 g Osmocote/day; Bruno et al. 2003: ~4.3 g Osmocote/day; Vega Thurber et al. 2009: ~5.4 g Osmocote/day), which reported increased seawater nutrient levels within the range of anthropogenic nutrient pollution (2–5× higher than ambient seawater: Bruno et al. 2003, 3× higher than ambient seawater: Szmant 2002, Vega Thurber et al. 2009). Though increased nutrient exposure may have resulted in effects on coral health or disease, the lack of precise measurements of nutrient-exposed seawater surrounding corals limits our interpretation of anthropogenic pollution effects on small acroporid corals.

While only predation affected coral success, we found that coral microbiomes strongly responded to both predation and nutrient enrichment. This study demonstrates that both bottom-up and top-down forces can be important regulators of coral-associated microbial communities, and may be for other foundation species and mutualistic microbes. Though both factors had significant effects, we found nutrient enrichment caused the most dramatic changes to coral microbiomes. These results are similar to studies in plant communities where altered species dominance occurred in response to nutrient enrichment (Tilman 1987, Levine et al. 1998). Nutrient loading has also been shown to alter mycorrhizal fungi that are closely associated with the health of terrestrial plants (Johnson et al. 2003). Determining the potential role of the coral microbiome in affecting coral health, disease, and fitness is similarly an important research frontier (Thompson et al. 2015). Importantly, in our study we did not observe any nutrient effects on corals despite strong changes to the composition of their microbiomes. As our study was conducted in a mid-shelf sandflat, future research should seek to compare microbial responses to multiple factors under natural coral reef conditions and over longer time periods.

The strong effects of nutrients on the microbiomes of *A. cervicornis* corals were largely driven by increases in *Rickettsia*-like organisms. *Rickettsiales* belong to a group of obligate intracellular parasites that are currently one of several suspected pathogens of white band disease (Miller et al. 2014*b*, Gignoux-Wolfsohn and Vollmer 2015), a disease that caused the near collapse of *A. cervicornis* and *A. palmata* populations throughout the Caribbean in the 1980s (Aronson and Precht 2001). As this disease continues to cause significant mortality for these coral species (Williams and Miller 2012), determining the factors controlling disease prevalence is critical for their recovery and the functioning of many Caribbean coral reef systems. Although nutrient loading has been positively correlated with coral disease prevalence in the Pacific (Haapkylä et al. 2011, Kaczmarsky and Richardson 2011) and found to increase disease prevalence in *Siderastrea* spp. corals over longer time periods (Vega Thurber et al. 2014), we found that only predation, and not nutrient enrichment, lead to coral mortality and signs of disease with *A. cervicornis*.

We did not find evidence of interactive effects between predation and nutrient enrichment on primary metrics of coral success or on coral microbial communities. Interestingly, we did find strong positive correlations between the relative abundances of several bacterial genera (Commamonadaceae, *Ralstonia*, and *Vibrio*) and tissue loss within snail + nutrient treatments, suggesting that these microbes may have been responding to both coral predation and nutrients. Our findings contrast with recent research showing that nutrient enrichment exacerbates the effects of coral predation leading to increased coral mortality (Zaneveld et al. 2016). This study, however, was conducted over 3 yr and with other coral species. Indeed, future research is required to determine whether these factors interact to affect endangered Caribbean acroporids.

Diverse ecosystems are increasingly under siege from anthropogenic stressors. While anthropogenic nutrient enrichment has been cited as a globally pervasive threat to many coastal and marine habitats (Lotze et al. 2006, Wear and Vega Thurber 2015), overfishing and climate change have increased populations of consumers that control critical foundation species including marsh grasses, kelps, trees, and coral reefs (reviewed in Silliman et al. 2013). Though these factors have interacted in other systems to facilitate disease or mortality, our results suggest that small corals may be highly susceptible to negative effects

caused by partial predation, but not nutrient enrichment, in the months following disturbance or restoration. While coral-associated microbial communities responded strongly to both biological and physical factors, the ultimate effects of these microbial changes on coral health remain to be examined.

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